Business Report – Predicting Customer Satisfaction from Review Text

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1 Executive Overview

1.1 Introduction

Today, customer satisfaction is no longer a nice-to-have option but a fundamental determinant of business success in all industries.

Satisfied customers tend to return, make recommendations, and positively impact the brand's reputation. Conversely, shared negative experiences of dissatisfied customers can harm the brand.

Currently, most companies, regardless of their industry, collect customer reviews through platforms such as Google, Amazon, Trustpilot, etc. These reviews provide extensive and valuable insights; however, most businesses lack the capacity to comprehensively analyze them on a large scale.

To address this challenge and undeniable need, we developed a machine learning system in this project, capable of automatically reading and analyzing the customer reviews. Our system predicts customer satisfaction level, which helps the company at two different levels:

1. Instant awareness: by detecting the dissatisfied and at-risk customers.
2. Deep insight: by revealing the causes that drive the customer to dissatisfaction or delight.

This way, the business is capable of making strategic decisions based on customers' comments without needing to manually read each one.

1.2 Project Vision

Our vision was to bridge the gap between data and understanding. We aimed to extract clear and useful insights from the unstructured customer review texts that the business could actually use.

To achieve this, we begin by building a model that can understand customer reviews in plain text form and predict the satisfaction level behind their words on a scale of 1 Star (completely dissatisfied) to 5 Stars (fully satisfied). Additionally, the model should be able to identify negative or at-risk customers immediately and provide the business with a clear explanation of the cause.

1.3 Key Results

We evaluated several methods, including traditional and modern machine learning models and obtained the following results:

- The **TF-IDF + Logistic Regression** model showed the best performance and interpretability, outperforming other complex neural approaches.
- For **multi-class prediction (1–5 stars)**, the model reached a **weighted F1 score of 0.7347**, indicating reliable accuracy across all satisfaction levels.

- For **binary classification (positive vs. negative)**, the system achieved an **exceptional F1 score of 0.9392**, correctly identifying over **92% of negative reviews**.
- The model was integrated and tracked through **MLflow,** assuring reproducibility and easy management.
- **SHAP explainability tools** were used to visualize the influence of certain words such as *excellent*, *love*, *easy* (positive) or *disappointed*, *late*, *poor* (negative) on model's predictions.

The final product of ours combines **high performance**, **transparency**, and **business applicability**, which make it an approachable and reliable tool for organizations seeking to understand their customers.

1.4 Business Value at a Glance

Business Challenge	Proposed Solution (ML-based)	Business Value
Too many customer reviews to read manually	Automatic review analysis using machine learning	Saves time and reduces labor costs
Late identification of dissatisfied customers	Real-time prediction of satisfaction levels	Enables faster reaction and customer recovery
Unclear reasons behind satisfaction trends	Transparent, word-based explanations	Provides clear insight into what drives happiness or frustration
Difficulty measuring improvements	Consistent, automated scoring system	Enables ongoing monitoring of customer satisfaction

1.5 Why This Matters

This system helps organizations collect and analyze customer feedback more efficiently. It offers a practical way to better understand what customers think.

The system reviews large amounts of real customer comments to find useful insights. These insights support better decisions, guide product development, and help the brand grow.

The project goes further than just predicting star ratings. It also finds and analyzes what matters most to customers and what concerns they have.

2 Business Context and Market Relevance

2.1 The Modern Challenge of Customer Feedback

Businesses often get a lot of customer feedback, but it can be hard to find clear insights in all the responses.

Customers share feedback after every purchase, hotel stay, or service. But this feedback is usually unstructured, inconsistent, and emotional, which makes it hard for computers to process.

A growing company cannot realistically review 1,000 customer comments each week, and larger organizations face even higher volumes. Manual review is slow, subjective, and expensive, even with a dedicated team.

On the other hand, neglecting customer feedback causes significant risks for any business:

- Dissatisfied customers frequently bypass direct complaints and instead share their experiences publicly online.
- Negative public reviews can hurt a company's reputation and sales.
- Managers may overlook emerging issues until they become significant problems.

Our project addresses these challenges directly by enabling companies to:

- identify dissatisfaction early.
- Detecting recurring topics and concerns.
- Monitor changes in customer satisfaction over time.
- Implement improvements informed by empirical data.

2.2 Strategic Importance

Customer satisfaction isn't just a quality metric; it's a powerful measure of how well a company is truly performing.

Researches consistently show that happy customers:

- Spend more on each purchase,
- Are more likely to recommend the brand to others, and
- Build long-term profitability through loyalty.

On the other hand, unhappy customers can quickly lead to higher churn rates and potential damage to a company's reputation.

By predicting satisfaction directly from customer reviews and comments, companies can:

- Turn emotional, unstructured feedback into measurable insights,

- Respond faster to emerging issues before they escalate online, and
- Understand what's driving sentiment — for example, how specific product changes affect customer perception.

The impact goes far beyond customer service. This kind of analysis also empowers smarter decisions in marketing, operations, and overall business strategy.

3 Project Objectives

3.1 Primary Goal

The main goal of our project was to build a reliable system that automatically predicts customer satisfaction levels based on written reviews.

In business terms, this means:

- Transforming open-ended text feedback into measurable satisfaction scores,
- Identifying early signs of dissatisfaction or potential customer loss, and
- Enabling better decisions through ongoing monitoring of customer sentiment.

3.2 Secondary Goals

While the primary goal was predictive performance, we also defined secondary objectives to ensure the system's real-world applicability:

- **Interpretability:**

The model should also be capable of reasoning the prediction. Meaning to clarify why a certain satisfaction level is predicted.

- **Efficiency:**

The system should process large volumes of data in a short time without excessive computational costs.

- **Scalability:**

The model should be adaptable to future extensions such as new languages, new data sources or other analytical layers.

- **Simplicity:**

The methodology should be simple enough to be understood and maintained by data teams, even without deep knowledge of deep learning.

- **Business readiness:**

The final product should be deployable. It should be easily integrated into existing business systems such as CRMs or dashboards.

3.3 Dual Modeling Approach

To address different business needs, our team designed two complementary prediction approaches:

Multi-Class Model	Predicts 1–5 satisfaction stars	Detailed satisfaction profile	Analytical insights, quality monitoring
Binary Model	Predicts Positive vs Negative sentiment	Fast alert system	Real-time customer service, reputation monitoring

This two-layer approach makes sure that both the big-picture decision-makers and the hands-on teams can work efficiently using the same system.

3.4 Success Criteria

We set clear performance and business goals to measure the project's success.

- **Accuracy:** The system needs to reliably predict satisfaction in each category.
- **Speed:** It should process and analyze reviews in real-time or as close to real-time as possible.
- **Interpretability:** Each prediction should be accompanied by a clear and easy-to-understand explanation.
- **Practicality:** The system should be simple to integrate, maintain, and scale in a real business setting.

The final solution met all these goals and even exceeded them in some areas.

4 Methodology and Approach

4.1 Overview of the Methodological Framework

Our primary goal was to develop a system that’s accurate, reliable, and easy to use — one that truly meets the needs of real businesses.

To achieve that, we followed a clear five-step data science process:

1. **Data Understanding and Preparation**
2. **Feature Engineering and Text Processing**
3. **Model Development and Training**

4. **Performance Evaluation and Optimization**

5. **Explainability and Business Validation**

We utilized **MLflow** to track and document every step, ensuring the entire process is transparent, well-organized, and easy to reproduce.

4.2 Data Understanding and Preparation

The dataset included thousands of customer reviews, each linked to a satisfaction rating from 1 to 5 stars.

Before jumping into modeling, we took time to really understand the data; exploring its structure, quirks, and potential challenges.

Here's what stood out:

- **Class imbalance:** Most reviews were positive (4–5 stars), while negative ones (1–2 stars) were much fewer.
- **Varied length:** Some reviews were just a few words; others stretched over several paragraphs.
- **Informal language:** Many included slangs, typos, emojis, or stretched words like "soooo good!!!".
- **Negations:** Phrases such as "not bad" or "not good" needed special handling to correctly interpret their sentiment.

To ensure our model was fair and accurate, we carefully cleaned and standardized all text data prior to training.

4.3 Data Cleaning and Preprocessing

The purpose of data cleaning is to remove unnecessary noise while preserving meaningful information.

Several steps were carefully applied in the following order:

- The whole text was lowercased to ensure uniformity.
- All special symbols, URLs, and HTML tags were removed.
- Contractions were expanded, for example, "can't" became "cannot," to improve clarity.
- All redundant whitespaces and punctuations were removed.
- The whole text was tokenized into individual words to support analysis.
- Negations were carefully handled to ensure that phrases like "not happy" would not be misread as positive.

This process significantly improved model consistency and interpretability.

4.4 Feature Engineering

Before feeding the data to the model, we extracted and engineered **text-based features** that represent how words appear and relate.

We experimented with three main types of feature extraction:

1. **TF-IDF (Term Frequency–Inverse Document Frequency)**
    - Measures how important a word is in a review compared to its frequency across all reviews.
    - Provides a strong baseline for classical models.
2. **Word Embeddings (Word2Vec, FastText)**
    - Converts words into numerical vectors that capture semantic meaning (e.g., *good* and *great* appear closer together).
3. **Transformer-based Embeddings (BERT / Sentence Transformers)**
    - Capture deeper context from sentences and surrounding words.

All three were tested under identical training conditions to ensure a fair comparison.

4.5 Handling Class Imbalance

Because satisfied customers tend to leave more reviews than dissatisfied ones, our data was **imbalanced**.

This imbalance can bias models to predict "positive" too often.

To correct this, we applied **resampling techniques**:

- **RandomOverSampler** to duplicate minority-class reviews.
- **SMOTE (Synthetic Minority Oversampling Technique)** to generate new synthetic examples for underrepresented ratings.

Balancing the data helped ensure that the model treated negative feedback as equally important.

4.6 Experiment Tracking with MLflow

All our experiments — from preprocessing configurations to model parameters and results — were tracked using **MLflow**.

This platform allowed us to:

- Record every model run and compare performance metrics.
- Visualize differences between versions.
- Reproduce any past experiment easily.

This rigorous documentation ensured the project's **transparency and auditability**, crucial for a professional defense.

5 Model Design and Evaluation

5.1 Overview of Models Tested

Several modeling strategies were tested in order to determine the best trade-off between **accuracy, speed, and interpretability.**

Model Type	Text Representation	Algorithms Tested	Best Performing Pair	Key Notes
Classical	TF-IDF	Logistic Regression, Linear SVC, LGBM, Random Forest	TF-IDF + Logistic Regression	Top overall performer (weighted F1 = 0.939 binary / 0.735 multi); fast, robust, interpretable.
Embedding-based	Word2Vec (Mean), FastText	Logistic Regression, LGBM	Word2Vec (Mean) + Logistic Regression	Captures semantic similarity but showed overfitting; lower generalization than TF-IDF.
Transformer-based	Sentence-BERT (all-MiniLM-L6-v2)	Logistic Regression, LGBM	*(Experiment incomplete)*	Strong potential for contextual understanding; computation-heavy. Future improvement area.

Three modeling families were evaluated across multi-class (1–5 stars) and binary (positive vs. negative) setups. The Classical TF-IDF + Logistic Regression model consistently delivered the highest and most reliable performance, outperforming more complex methods.

Embedding-based approaches (Word2Vec, FastText) provided richer semantic representations but failed to retain the intensity of sentiment, revealing that the primary limitation stemmed from feature construction rather than model complexity.

Ultimately, a dual-model strategy was adopted:

The multi-class **TF-IDF + Logistic Regression** model serves as an analytical tool for examining neutral or mid-range reviews.

The binary **TF-IDF + Logistic Regression** model is ready for production use, achieving excellent accuracy and a high recall rate for negative reviews, making it ideal for real-time sentiment monitoring.

5.2 Why Logistic Regression?

Although neural networks and transformers are often considered the gold standard of modern AI, our experiments demonstrated that well-tuned classical models can perform exceptionally well.

The **Logistic Regression** model paired with **TF-IDF** features consistently delivered:
• The highest F1 scores across both tasks.
• The fastest training and prediction times.
• The greatest interpretability, as confirmed through SHAP analysis.

Beyond its strong performance, Logistic Regression also proved to be the **most stable model**, maintaining minimal performance gaps between training and test sets. This balance of **accuracy and generalization** distinguishes it from more complex models such as LinearSVC and LGBM, which achieved slightly higher raw scores but exhibited pronounced overfitting.

This combination of **predictive strength, efficiency, and robustness** makes the **TF-IDF + Logistic Regression** approach the most practical and trustworthy choice for real-world business deployment.

5.3 Evaluation Metrics

The model's reliability was assessed using several important performance metrics:

- **F1 Score:** Measures how well the model balances precision and recall, ensuring it identifies the correct results effectively.
- **Weighted F1:** Provides a fair evaluation when some categories have many more samples than others.
- **Recall for the Negative Class:** This is especially critical for spotting dissatisfied customers early.

Together, these metrics give a clear picture of the model's overall accuracy and its ability to detect early warning signs of customer dissatisfaction.

5.4 Multi-Class Model Results (1-5 Stars)

The multi-class model performed strongly across all satisfaction levels, achieving a **weighted F1 score of 0.7347**.

It successfully captured clear differences between:

- **5-star reviews** (highly positive language, e.g., "amazing," "fast," "excellent").
- **1-star reviews** (strongly negative tone, e.g., "terrible," "broken," "never again").

Predicting middle categories (2–4 stars) was more challenging, as these reviews often mix both praise and complaints (e.g., “good product, but delivery was slow”).

Still, overall results demonstrated robust classification with valuable granularity.

5.5 Binary Model Results (Positive vs. Negative)

When we simplified the problem into two categories — **positive (4–5 stars)** and **negative (1–2 stars)** — the model achieved outstanding performance:

- **Weighted F1 Score:** 0.9392
- **Recall (Negative Class):** 92.1%

This means the model correctly identifies over 90% of unhappy customers.

In business terms, it’s an **extremely reliable early-warning system.**

5.6 Hyperparameter Optimization with Optuna

We used **Optuna**, an advanced optimization framework, to fine-tune the key hyperparameters of each model configuration

By automatically exploring hundreds of parameter combinations, Optuna identified the best-performing setup. This systematic tuning process led to consistent and measurable improvements across all experiments.

5.7 Explainability with SHAP

Interpretability is essential for real-world business use because people need to understand *why* the model makes specific predictions.

We used **SHAP (Shapley Additive Explanations)** to visualize how individual words influenced the model’s outcomes. This approach enabled us to:

- Identify which terms were most strongly linked to positive or negative satisfaction.
- Provide clear, transparent explanations to business stakeholders about each prediction.

For example:

- **Positive indicators:** *excellent, love, easy, amazing*
- **Negative indicators:** *disappointed, broken, slow, refund*

These insights made the model’s behavior **explainable and trustworthy**, ensuring alignment between technical soundness and strategic business goals.

6 Key Findings and Insights

6.1 Summary of Quantitative Results

Text Representation + Model	Task Type	Weighted F1 Score	Recall	Interpretability	Speed
TF-IDF + Logistic Regression	Multi-Class	0.7347	-	High	Fast
TF-IDF + Logistic Regression	Binary	0.9392	92.1%	High	Very Fast

6.2 Main Takeaways

- **Simplicity Over Complexity**
  When the data was carefully prepared, the classical models really stood out. It turns out that a well-tuned, simple approach can often perform just as well — or even better — than more complicated ones.

- **Efficiency and Clarity**
  The **TF-IDF + Logistic Regression** model proved to be both quick and easy to understand, striking an ideal balance between performance and transparency.

- **Accuracy Where It Counts**
  With a **92% recall** for negative reviews, the binary model gives businesses a dependable way to catch early signs of customer dissatisfaction before issues escalate.

- **Trust Through Transparency**
  Thanks to **SHAP visualizations**, stakeholders can see exactly why the model flagged certain reviews. This level of clarity makes AI insights easier to trust, explain, and act upon.

- **Good Data Makes Great Models**
  One thing became clear throughout the project — strong results start with clean data. Careful preprocessing played a major role in keeping the model's performance stable and reliable.

6.3 Business Interpretation of Results

In practical business terms, our model can:

- Analyze thousands of customer reviews within seconds.

- Automatically flag reviews that indicate potential problems.

- Offer clear, easy-to-understand explanations for each prediction.
- Help managers respond faster and make better-informed decisions.

In short, the system isn't just accurate; it's genuinely useful and ready to drive real action.

7 Business Applications and Use Cases

7.1 Turning Data into Decision

The real value of an analytical model isn't in how complex it is, but in the quality of decisions it helps people make.

Our satisfaction prediction model turns customer feedback from something that's merely collected into something that actively drives decisions.

By automatically analyzing reviews, the model helps companies:

- Spot early signs of dissatisfaction and prevent customer loss.
- Understand what truly influences satisfaction across their services.
- Monitor how customer perception shifts after new initiatives.
- Use data to guide improvements in service, marketing, and operations.

In essence, it changes the way businesses listen to their customers — moving from occasional surveys to continuous, large-scale understanding.

7.2 Core Use Cases

We identified four main areas where our system can bring immediate, tangible business benefits:

1. Real-Time Alert System for Customer Service

- Every new review can be automatically analyzed and scored.
- When the model detects negative sentiment, it sends an alert to the customer service team, prompting immediate follow-up.
- This reduces the time between dissatisfaction and resolution; preventing escalation and protecting brand reputation.

Business impact: Faster problem resolution → higher retention → fewer public complaints.

2. Customer Experience Dashboard

- The model's results can be summarized into an interactive dashboard, providing managers with real-time insights.

- Metrics such as overall satisfaction trend, top complaint topics, and frequent positive terms can be visualized clearly.

Business impact: Data-driven decision-making and better prioritization of improvement efforts.

3. Marketing and Product Development Insights

By analyzing which words most frequently appear in positive or negative reviews, the company gains a deeper understanding of what customers value most.

For example:

- If “fast delivery” often appears in positive reviews, logistics excellence is a strength to promote.
- If “price” appears in negative reviews, marketing teams know where to adjust communication or pricing strategy.

Business impact: Aligns marketing messages with what truly drives customer satisfaction.

4. Quality and Operations Monitoring

- The model can be used internally to detect recurring patterns in negative feedback.

For example, repeated complaints about “broken packaging” or “slow customer support” signal process weaknesses that require operational intervention.

Business impact: Enables early detection of systemic issues and supports continuous improvement.

7.3 Extended Opportunities

The same architecture could later support other text-based business analyses, such as:

- Employee feedback surveys (for HR insights).
- Social media monitoring (brand reputation tracking).
- Open-ended responses in market research.

This makes the solution **scalable and useful for many purposes, rather than** just a one-off model.

8 Implementation and Integration

8.1 Deployment Overview

For an analytical solution to truly create value, it needs to fit seamlessly into a company’s everyday operations.

That's why our model was built with that practicality in mind. It can easily integrate with existing systems through standard data pipelines or APIs, ensuring a smooth and efficient adoption process.

8.2 System Workflow

The operational workflow consists of four main steps:

1. **Data Input:** Customer reviews are collected automatically from the company's platforms (website, surveys, or social media).
2. **Model Processing:** The review text is cleaned, analyzed, and classified by the satisfaction prediction model.
3. **Output Generation:** Each review is assigned a satisfaction score (1–5 or Positive/Negative).
4. **Integration:** The results are exported to a dashboard, CRM, or notification system.

8.3 Technical Simplicity and Maintenance

Although the underlying concept involves machine learning, the deployed solution remains simple to operate:

- It can run on standard computing infrastructure.
- It processes thousands of reviews within minutes.
- It requires minimal human supervision beyond occasional retraining.

This ensures **low maintenance costs** and long-term sustainability.

8.4 Integration Scenarios

Possible integration approaches include:

- **CRM systems** can automatically identify dissatisfied customers, enabling support teams to respond promptly and prevent issues from escalating.
- **Business Intelligence (BI) dashboards** can combine satisfaction scores and sentiment trends with existing sales and operational KPIs, offering a clear and comprehensive view of overall performance.
- **Automated email alerts** can immediately notify the relevant teams when negative reviews are detected, ensuring timely and consistent responses.

This flexibility allows the model to integrate seamlessly into each organization's existing ecosystem, making implementation quick and straightforward.

8.5 Monitoring and Continuous Improvement

Like any business process, the model needs regular monitoring and updates to stay accurate and relevant.

Key actions include:

- Tracking core performance metrics over time.
- Reviewing SHAP explanations to spot new or changing themes in customer feedback.
- Retraining the model periodically—ideally every quarter or whenever new data patterns emerge.

These steps help ensure the system continues to reflect real customer behavior and delivers reliable insights.

9 Business Value Analysis

9.1 Measurable Business Impact

Our system consistently delivers real, measurable improvements across key areas of the business.

Business Dimension	***Key Contribution***	***Measured Impact***
Operational Efficiency	Automates manual review reading	Estimated 80–90% reduction in human review workload
Customer Retention	Identifies at-risk customers early	Up to 25% improvement in retention rate (depending on response actions)
Decision-Making Quality	Turns qualitative data into clear, actionable metrics	Enables evidence-based management
Brand Reputation	Reduces negative public exposure	Improves online ratings over time
Cost Efficiency	Uses lightweight models requiring minimal infrastructure	Significantly lower cost than deep-learning alternatives

Note: The figures shown (e.g., Operational Efficiency, Customer Retention) are based on industry benchmarks and prior studies in automation and customer analytics. They illustrate **potential business gains rather than direct results measured in this project.*

9.2 Qualitative Benefits

Beyond numbers, the system also creates strategic and cultural benefits:

- Encourages a data-driven culture across departments.

- Gives employees a clear picture of customer needs.
- Strengthens accountability and transparency in customer experience management.
- Builds trust, since insights are based on what customers actually say, not on assumptions.

This shift from intuition to evidence marks a profound organizational improvement.

9.3 Value for Different Stakeholders

Customer Service Teams	Get early alerts for negative feedback and can react quickly.
Marketing Department	Understand what drives positive sentiment and tailor campaigns.
Product Development	Identify design or quality issues directly from user feedback.
Top Management	Monitor satisfaction trends as a KPI for company performance.

10 Limitations and Future Outlook

10.1 Recognized Limitations

While the system performs well, we recognize it is not without limitations.

- **Middle-Range Ambiguity (2–4 Stars):**
  Mid-range reviews often combine positive and negative feedback, such as *"Good product, but delivery was slow."* These are more challenging for the model to classify accurately. Future models should be more context-aware to better address this matter.

- **Language Limitation:**
  Currently, the model processes only English reviews. Expanding support to additional languages, such as German or French, would increase its value for international businesses.

- **Computational Balance:**
  The simple models we use are fast and efficient, but they may overlook details that advanced transformer models can identify. Combining both approaches could offer an optimal solution in the future.

- **Transformer Experiment Constraint:**
  The transformer-based experiment was not completed due to computational bottlenecks during the hyperparameter optimization phase. Each optimization trial required regenerating embeddings for the full dataset, leading to prohibitive runtimes. Future work should adopt a pre-computed embedding workflow to make transformer experimentation feasible.

- **Changing Vocabulary:**
  Language evolves continuously, with new slang, emojis, and expressions emerging regularly. To maintain system accuracy, ongoing retraining and vocabulary updates will be necessary.

10.2 Potential Future Enhancements

Building on this strong foundation, we see plenty of room to make the system even smarter and more useful for businesses:

- **Multilingual Expansion:** Add support for multiple languages so global companies can analyze reviews from different markets.
- **Topic Detection:** Automatically spot key themes like *delivery*, *price*, or *customer support* to uncover the real reasons behind satisfaction or frustration.
- **Sentiment Trend Monitoring:** Track how customer sentiment changes over time across regions, products, or services.
- **Transformer Integration:** Explore newer models like BERT or RoBERTa when more computing power is available, to capture deeper context in customer feedback.
- **Automation Feedback Loop:** Feed the model's results directly back into business systems, so teams can react and make improvements faster.

10.3 Long-Term Vision

Our long-term vision is to help organizations build customer-centric ecosystems powered by real-time understanding of satisfaction.
In such a system, customer feedback would not just sit in a database. Instead, it would actively guide daily business decisions.
This project demonstrates how artificial intelligence can serve human purposes, including understanding, empathy, and improved service.

11 Lessons Learned

11.1 Key Insights from the Project

As we carried out this Project, we gained a clear understanding of what actually makes a data-driven solution function efficiently in a business setting. It became clear that keeping things simple, maintaining high data quality, and making results easy to understand matter far more than building overly complex models. The most effective approach was to follow a well-organized process, use clear reasoning at every step, and prioritize efficient execution. These principles apply across any industry.

11.2 Challenges and Responses

As with any real project, we faced a few bumps along the way. One big challenge was dealing with imbalanced data, which we handled using resampling methods. We also had to work within computational limits, which prompted us to choose lighter, faster models over heavier ones. Looking back, these choices weren't compromises; they reflected a practical business mindset focused on speed, value, and sustainability rather than experimentation for its own sake.

11.3 Takeaway

Ultimately, we discovered that clarity, transparency, and teamwork are the essential keys to success in applied data science. When people can understand and trust what a model is doing, it becomes more than just a technical tool; it turns into something that genuinely supports better decisions.

12 Business Deployment Recommendation

12.1 Recommended Strategy

After reviewing all our experiments, we found that the best approach is to use a dual-model setup.

Model Type	Purpose	Key Advantages
Binary Model (TF-IDF + Logistic Regression)	Real-time alerts for negative feedback	High speed, 94% F1 score, enables immediate action
Multi-Class Model (TF-IDF + Logistic Regression)	Detailed satisfaction analysis (1–5 stars)	Provides granular insights and supports long-term trend tracking

Both approaches use the same preprocessing pipeline, which makes integration smooth and keeps maintenance simple.

12.2 Operational Recommendation

We recommend that businesses use this system in two complementary layers:

- **Layer 1 – Operational Monitoring:**
  - The binary model runs continuously, scanning all new reviews.

- When negative sentiment is detected, an automatic alert is sent to the responsible department.

- **Layer 2 – Strategic Insight:**
  - The multi-class model runs periodically to analyze large batches of reviews.
  - Its results are summarized in dashboards for quarterly business reviews and product evaluations.

12.3 Risk Management and Mitigation

Risk	Potential Impact	Mitigation Strategy
Biased or Unbalanced Input Data	Misleading results	Conduct regular data audits and retrain the model periodically
Misinterpretation of Results	Incorrect or uninformed business decisions	Provide clear documentation and staff training
Data Privacy Concerns	Legal or ethical risks	Ensure full GDPR compliance and apply data anonymization
Technical Maintenance Issues	Model performance degradation over time	Implement scheduled monitoring and maintenance routines

13 Ethical and Practical Considerations

13.1 Transparency and Explainability

AI systems must be transparent, especially when their output influences business decisions. Our model's interpretability through SHAP analysis ensures that users understand why a review was classified as positive or negative.

13.2 Data Privacy and Compliance

All data in this project was handled with care to protect privacy. Personal details were removed, and the model based its predictions only on the text of the reviews. Although this wasn't a formal GDPR compliance audit, we made sure to minimize privacy risks and treat all data responsibly.

13.3 Human Oversight

The model is designed to assist, not replace human judgment.
Customer service representatives should always review critical alerts before taking action.
This hybrid approach, combining machine speed with human empathy, leads to the best business outcomes.

14 Future Outlook

14.1 Next Steps for Development

Our results and what we learned have made the next development steps clear and organized.
Before expanding the system's capabilities, the first priority is to strengthen the core model. This means improving the Transformer experiment with a pre-computed workflow, training domain-specific embeddings on our review data, and examining misclassified reviews to better handle mixed sentiments. These improvements will build a stronger foundation for a more scalable and intelligent solution.
After refining the core model, we can develop the system further to boost its business value and usability:

- **Multilingual Expansion:** Add language detection and translation to make the system work for global markets.
- **Topic Detection Module:** Automatically sort reviews by key themes like *price*, *delivery*, or *quality* to provide more focused insights.
- **Real-Time API Service:** Make the model available as a cloud-based API to easily connect with current business systems.
- **Advanced Visualization Dashboard:** Create interactive dashboards with BI tools like Power BI, Tableau, or Dash to make results easy to explore and use.
- **Hybrid Transformer Approach:** Use lightweight Transformer models like DistilBERT when more computing power is available. This combines better language understanding with efficient performance.

15 Conclusion

15.1 Final Summary

Our project, *Predicting Customer Satisfaction from Review Text*, successfully demonstrated that **machine learning can transform unstructured customer feedback into actionable insight**.

Through disciplined methodology, careful experimentation, and a focus on clarity, our team — **Fred Richter**, **Alex Pötzl**, and **Yalda Khoshhal** — delivered a system that:

- Accurately predicts satisfaction levels (F1 up to 0.94).
- Explains its reasoning transparently.
- Operates efficiently on standard infrastructure.
- Provides measurable value for businesses.

We showed that **artificial intelligence can be both powerful and understandable**, offering a bridge between complex data and everyday business decisions.

15.2 Strategic Impact

With this model in place, businesses can stay closely connected to their customers in real time. They can track satisfaction continuously, respond to negative feedback before it escalates, understand what truly drives customer feelings, and make better, evidence-based decisions.

What we've built is more than a technical system; it's a practical partner for customer understanding and excellence.

15.3 Closing Statement

This project was guided by a simple belief: technology should always work in service of people.

By teaching a machine to read and interpret human language, we've created more than just a model; we've built a way for organizations to listen with greater empathy, act with more insight, and strengthen the relationships that matter most.